Program 1:

```
# ---------------------------------------------------------------------
# SHOPPING CART SYSTEM WITH CATEGORIES
# ---------------------------------------------------------------------
# Constants
DISCOUNT_RATE = 0.10       # 10% discount
GST_RATE = 0.05            # 5% GST

subtotal = 0               # To store total amount
cart_details = ""          # To display full bill

print("================ SHOPPING CART SYSTEM ================")
print("Categories available:")
print("1. Grocery")
print("2. Electronics")
print("3. Clothing")

# Get number of items
num_items = int(input("\nEnter how many items you want to add: "))

# ------------------------- LOOP START -------------------------
for i in range(1, num_items + 1):

    print(f"\nEnter details for Item {i}:")

    # Category selection
    category_choice = int(input("Choose category (1-Grocery, 2-Electronics, 3-Clothing): "))

    if category_choice == 1:
        category = "Grocery"
    elif category_choice == 2:
        category = "Electronics"
    elif category_choice == 3:
        category = "Clothing"
    else:
        category = "Others"

    # Item details
    item_name = input("Item Name: ")
    item_price = float(input("Item Price: "))
    item_quantity = int(input("Quantity: "))

    # Calculate item total
    item_total = item_price * item_quantity

    # Add to subtotal
    subtotal = subtotal + item_total

    # Store details for bill display
    cart_details = cart_details + (
        f"\nItem {i}:\n"
        f"  Category : {category}\n"
        f"  Name     : {item_name}\n"
        f"  Price    : {item_price}\n"
```

```
        f"  Quantity : {item_quantity}\n"
        f"  Total    : {item_total}\n"
    )

# ------------------------- BILL CALCULATION -------------------------

discount_amount = subtotal * DISCOUNT_RATE
after_discount = subtotal - discount_amount
gst_amount = after_discount * GST_RATE
final_total = after_discount + gst_amount

# ------------------------- BILL OUTPUT -------------------------

print("\n=====================================================")
print("                    FINAL BILL                    ")
print("=====================================================")
print(cart_details)
print("-----------------------------------------------------")
print("Subtotal:", subtotal)
print("Discount (10%):", discount_amount)
print("After Discount:", after_discount)
print("GST (5%):", gst_amount)
print("-----------------------------------------------------")
print("FINAL BILL AMOUNT:", final_total)
print("=====================================================")
print("             THANK YOU FOR SHOPPING!              ")
print("=====================================================")
```

Program 2:

```
# Fitness Membership Recommendation System
# Demonstrates conditional & decision control statements

print("=== Welcome to FitLife Membership Advisor ===")

# Taking user inputs
age = int(input("Enter your age: "))
goal = input("Enter your fitness goal (weightloss/muscle/cardio/general): ").lower()
budget = int(input("Enter your monthly budget (in ₹): "))

print("\nAnalyzing your inputs...\n")

# Decision Control Statements
if age < 18:
    print("You qualify only for the Teen Fitness Plan.")
    if budget >= 800:
        print("Recommended Plan: Teen Basic - ₹800/month")
    else:
        print("Budget is too low for any plan. Consider home workouts.")
else:
    # Age 18 and above
    if goal == "weightloss":
        if budget >= 2000:
            print("Recommended Plan: Weight Loss Pro (Zumba + Diet Plan + Gym) - ₹2000/month")
        elif budget >= 1200:
            print("Recommended Plan: Weight Loss Basic (Cardio + Trainer) - ₹1200/month")
        else:
            print("Budget too low. Try walking/jogging daily.")

    elif goal == "muscle":
        if budget >= 2500:
            print("Recommended Plan: Muscle Gain Ultimate (Heavy Gym + Personal Trainer) - ₹2500/month")
        elif budget >= 1500:
            print("Recommended Plan: Muscle Gain Basic - ₹1500/month")
        else:
            print("Budget too low. Try calisthenics at home.")

    elif goal == "cardio":
        if budget >= 1000:
            print("Recommended Plan: Cardio Unlimited (Treadmill + Cycling) - ₹1000/month")
        else:
            print("Budget too low for cardio plan. Try skipping rope.")

    elif goal == "general":
        if budget >= 800:
            print("Recommended Plan: General Fitness Plan - ₹800/month")
        else:
            print("Budget too low. Consider home exercises.")
    else:
        print("Unknown goal entered. Please try again.")

print("\nThank you for using FitLife Membership Advisor!")
```

```
#### Initialize variables
books_available = ["Python", "Database Management", "Operating Systems", "Computer
Networks", "Open Source Tools"]
books_taken = []
books_returned = []
books_to_return = []
# Main loop
while True:
    print("\n==== Library Management System ====")
    print("1. Display available books")
    print("2. Issue a book")
    print("3. Return a book")
    print("4. Display books to be returned")
    print("5. Exit")

    choice = input("Enter your choice: ")
    if choice == "1":
        print("\nAvailable Books:")
        for book in books_available:
            print("-", book)
    elif choice == "2":
        print("\nIssue a Book:")
        book_name = input("Enter the book name to issue: ")
        if book_name in books_available:
            books_available.remove(book_name)
            books_taken.append(book_name)
            print(f"{book_name} book has been issued successfully.")
        else:
            print("Sorry, the book is not available.")

    elif choice == "3":
        print("\nReturn a Book:")
        book_name = input("Enter the book name to return: ")
        if book_name in books_taken:
            books_taken.remove(book_name)
            books_available.append(book_name)
            books_returned.append(book_name)
            print(f"{book_name}  book has been returned successfully.")
        else:
            print("You have not taken this book.")

    elif choice == "4":
        print("\nBooks to be Returned:")
        for book in books_taken:
            if book not in books_returned:
                books_to_return.append(book)
                print("-", book)
        if not books_to_return:
            print("No books currently need to be returned.")
```

```
elif choice == "5":
    print("Exiting Library Management System!")
    break

else:
    print("Invalid choice! Please enter a valid option.")
```

4. a. Write a Python program to demonstrate the use of functions and modules.

 # cab_module.py

```
# Function to show cab types
def show_cab_types():
    print("\nAvailable Cab Types:")
    print("1. Mini        - ₹10 per km")
    print("2. Sedan       - ₹15 per km")
    print("3. SUV         - ₹20 per km")

# Function to calculate fare
def calculate_fare(cab_choice, distance):
    if cab_choice == 1:
        return distance * 10
    elif cab_choice == 2:
        return distance * 15
    elif cab_choice == 3:
        return distance * 20

# Function to get cab name
def get_cab_name(cab_choice):
    if cab_choice == 1:
        return "Mini"
    elif cab_choice == 2:
        return "Sedan"
    elif cab_choice == 3:
        return "SUV"

# Function to print booking receipt
def print_receipt(name, pickup, drop, cab_name, distance, total_amount):
    print("\n=========== CAB BOOKING RECEIPT ===========")
    print(f"Passenger Name : {name}")
    print(f"Pickup Point   : {pickup}")
    print(f"Drop Point     : {drop}")
    print(f"Cab Type       : {cab_name}")
    print(f"Distance       : {distance} km")
    print(f"Total Fare     : ₹{total_amount}")
    print("Status         : BOOKED SUCCESSFULLY")
    print("==========================================")
```

 # main.py

```
import cab_module as cab

print("=========== CITY CAB BOOKING SYSTEM ===========")

name = input("Enter your name: ")
pickup = input("Enter pickup location: ")
drop = input("Enter drop location: ")
distance = float(input("Enter distance between locations (in km): "))

cab.show_cab_types()
cab_choice = int(input("\nSelect your cab type: "))

# Get cab name
cab_name = cab.get_cab_name(cab_choice)

# Calculate fare
amount = cab.calculate_fare(cab_choice, distance)

# Print final booking receipt
cab.print_receipt(name, pickup, drop, cab_name, distance, amount)
```

4. b. Write a Python program to demonstrate the use of functions and modules.

 # movie_module.py
 # Demonstrating Functions + Modules

```
# Function to display movies
def show_movies():
    print("\nAvailable Movies:")
    print("1. Christmas Karma")
    print("2. My Secret Santa")
    print("3. Oh. What. Fun.")

# Function to show available timings
def show_timings():
    print("\nShow Timings:")
    print("1. 10:00 AM")
    print("2. 2:00 PM")
    print("3. 6:00 PM")
    print("4. 9:00 PM")

# Function to show ticket types
def show_ticket_types():
    print("\nTicket Categories:")
    print("1. Regular  - ₹120")
    print("2. Premium  - ₹180")
    print("3. VIP      - ₹250")

# Function to return ticket price
def get_ticket_price(type_choice):
    if type_choice == 1:
        return 120
    elif type_choice == 2:
        return 180
    elif type_choice == 3:
        return 250

# Function to generate final booking receipt
def print_receipt(name, movie, timing, ticket_type, seats, total_amount):
    print("\n================ BOOKING RECEIPT ================")
    print(f"Name          : {name}")
    print(f"Movie         : {movie}")
    print(f"Show Time     : {timing}")
    print(f"Ticket Type   : {ticket_type}")
```

```
    print(f"No. of Seats   : {seats}")
    print(f"Total Amount   : ₹{total_amount}")
    print("Booking Status : CONFIRMED")
    print("==================================================")
```

 # main.py → Movie Ticket Booking Application

```
import movie_module as mv
print("========= Welcome Phoenix Mall Of Asia Bengaluru =========")
name = input("Enter your name: ")

# ----------------- MOVIE SELECTION -----------------
mv.show_movies()
movie_choice = int(input("Select a movie: "))
movie_names = ["Christmas Karma", "My Secret Santa", "Oh. What. Fun."]
selected_movie = movie_names[movie_choice - 1]

# ----------------- TIMING SELECTION ----------------
mv.show_timings()
time_choice = int(input("Select show time: "))
timings = ["10:00 AM", "2:00 PM", "6:00 PM", "9:00 PM"]
selected_time = timings[time_choice - 1]

# ----------------- TICKET TYPE SELECTION -----------
mv.show_ticket_types()
ticket_choice = int(input("Select ticket type: "))
ticket_types = ["Regular", "Premium", "VIP"]
ticket_type = ticket_types[ticket_choice - 1]

# ----------------- SEAT COUNT ----------------------
seats = int(input("Enter number of seats: "))

# Ticket price from module function
price = mv.get_ticket_price(ticket_choice)
total = price * seats

# ----------------- PRINT RECEIPT -------------------
mv.print_receipt(name, selected_movie, selected_time, ticket_type, seats, total)
```

```
# ---------------- EXCEPTION HANDLING----------------
class InvalidAgeError(Exception):
    pass

while True:
    print("\n=========== ONLINE SERVICE MENU ===========")
    print("1. ATM Withdrawal (ZeroDivisionError)")
    print("2. Mobile Recharge Amount (ValueError)")
    print("3. Product Selection (IndexError)")
    print("4. Opening Invoice File (FileNotFoundError)")
    print("5. Login System (Multiple Exceptions)")
    print("6. Ticket Booking (try-except-else)")
    print("7. Payment Gateway (try-except-finally)")
    print("8. Age Verification (User-Defined Exception)")
    print("9. Exit")

    try:
        choice = int(input("Enter your choice: "))
    except ValueError:
        print("Please enter a valid menu number!")
        continue

    # ------------------------------------------------
    if choice == 1:
        print("\n--- ATM Withdrawal System ---")
        try:
            balance = int(input("Enter account balance: "))
            users = int(input("Enter number of users sharing account: "))
            amount_per_user = balance / users
            print("Amount per user:", amount_per_user)
        except ZeroDivisionError:
            print("Error: Number of users cannot be zero!")

    # ------------------------------------------------
    elif choice == 2:
        print("\n--- Mobile Recharge System ---")
        try:
            amount = int(input("Enter recharge amount: "))
            print("Recharge of ₹", amount, "successful!")
        except ValueError:
            print("Error: Recharge amount must be a number!")

    # ------------------------------------------------
    elif choice == 3:
```

```
    print("\n--- Online Shopping Product Selection ---")
    products = ["Mobile", "Laptop", "Headphones"]
    print("Available products:", products)
    try:
        index = int(input("Select product index: "))
        print("You selected:", products[index])
    except IndexError:
        print("Error: Invalid product selection!")
    except ValueError:
        print("Error: Enter a valid number!")

# ---------------------------------------------------
elif choice == 4:
    print("\n--- Invoice Download System ---")
    try:
        file = open("invoice.txt", "r")
        print(file.read())
        file.close()
    except FileNotFoundError:
        print("Error: Invoice file not found!")

# ---------------------------------------------------
elif choice == 5:
    print("\n--- User Login System ---")
    try:
        username = input("Enter username: ")
        password = int(input("Enter numeric password: "))
        print("Login successful!")
    except ValueError:
        print("Error: Password must be numeric!")
    except Exception:
        print("Error: Login failed due to unknown issue!")

# ---------------------------------------------------
elif choice == 6:
    print("\n--- Movie Ticket Booking ---")
    try:
        seats = int(input("Enter number of seats: "))
        print("Seats booked:", seats)
    except ValueError:
        print("Error: Seat count must be numeric!")
    else:
        print("Booking confirmed successfully!")
```

```
# ---------------------------------------------------
elif choice == 7:
    print("\n--- Online Payment Gateway ---")
    try:
        amount = int(input("Enter payment amount: "))
        print("Payment of ₹", amount, "processed")
    except ValueError:
        print("Error: Invalid payment amount!")
    finally:
        print("Transaction session closed.")

# ---------------------------------------------------
elif choice == 8:
    print("\n--- Age Verification for Service ---")
    try:
        age = int(input("Enter your age: "))
        if age < 18:
            raise InvalidAgeError
        print("Access granted!")
    except InvalidAgeError:
        print("Error: Age must be 18 or above!")
    except ValueError:
        print("Error: Age must be numeric!")

# ---------------------------------------------------
elif choice == 9:
    print("\nThank you for using the Online Service System!")
    break

else:
    print("Invalid option! Please choose again.")
```

5. Implement operations on lists in Python.

```
# Grocery Shopping List

# Initialize the grocery list
grocery_list = ["rice", "pasta", "bread", "oats"]

def display_menu():
    print("\n====== Grocery Shopping List Menu ======")
    print("1. Display list")
    print("2. Add item")
    print("3. Insert item at a position")
    print("4. Remove item")
    print("5. Update/Replace item")
    print("6. Search for item")
    print("7. Sort list (A-Z)")
    print("8. Reverse list")
    print("9. Exit")
    print("========================================")
display_menu()
while True:

    choice = input("Enter your choice (1-9): ")
    if choice == '1':
        if grocery_list:
            print("Current Grocery List:", grocery_list)
        else:
            print("The grocery list is empty.")

    elif choice == '2':
        item = input("Enter item to add: ")
        grocery_list.append(item)
        print(f"'{item}' added to the list.")
        print("Grocery List After Adding :", grocery_list)
        print("==============================")

    elif choice == '3':
        item = input("Enter item to insert: ")
        pos = int(input("Enter position (starting from 0): "))
        if 0 <= pos <= len(grocery_list):
            grocery_list.insert(pos, item)
            print(f"'{item}' inserted at position {pos}.")
            print("Grocery List after Inserting:", grocery_list)
            print("==============================")
        else:
```

```
        print("Invalid position!")

elif choice == '4':
    item = input("Enter item to remove: ")
    if item in grocery_list:
        grocery_list.remove(item)
        print(f"'{item}' removed from the list.")
        print("Grocery List After Removing:", grocery_list)
    else:
        print(f"'{item}' not found in the list.")

elif choice == '5':
    old_item = input("Enter item to replace: ")
    if old_item in grocery_list:
        new_item = input("Enter new item: ")
        index = grocery_list.index(old_item)
        grocery_list[index] = new_item
        print(f"'{old_item}' replaced with '{new_item}'.")
        print("Grocery List after Replacing:", grocery_list)
    else:
        print(f"'{old_item}' not found in the list.")

elif choice == '6':
    item = input("Enter item to search: ")
    if item in grocery_list:
        print(f"'{item}' found at index {grocery_list.index(item)}.")

    else:
        print(f"'{item}' not found in the list.")

elif choice == '7':
    grocery_list.sort()
    print("List sorted in alphabetical order.")
    print("Sorted Grocery List :", grocery_list)

elif choice == '8':
    grocery_list.reverse()
    print("List order reversed.")
    print("Reversed Grocery List:", grocery_list)

elif choice == '9':
    print("Exiting program. Happy shopping!")
    break

else:
    print("Invalid choice! Please enter 1-9.")
```

```python
# Weekly Class Schedule - Tuple Operations

# Initial tuple
schedule = ("datastructures", "networks", "operatingsystems", "python", "java")
def display_menu():
    print("\n====== Tuple Operations Menu ======")
    print("1. Display schedule")
    print("2. Length of tuple")
    print("3. Access subject by index")
    print("4. Search subject")
    print("5. Count subject occurrences")
    print("6. Get index of subject")
    print("7. Slice the tuple")
    print("8. Sort subjects (A-Z)")
    print("9. Add a subject (convert to list)")
    print("10. Find max & min (alphabetical)")
    print("11. Exit")
    print("==================================")

while True:
    display_menu()
    choice = input("Enter your choice (1-11): ")
    if choice == '1':
        print("Weekly Schedule:", schedule)
    elif choice == '2':
        print("Total subjects:", len(schedule))
    elif choice == '3':
        index = int(input("Enter index: "))
        if 0 <= index < len(schedule):
            print("Subject:", schedule[index])
```

```
    else:
        print("Invalid index")
elif choice == '4':
    subject = input("Enter subject to search: ")
    if subject in schedule:
        print(f"{subject} is in the schedule")
    else:
        print(f"{subject} not found")
elif choice == '5':
    subject = input("Enter subject to count: ")
    print(f"Occurrences of {subject}: {schedule.count(subject)}")
elif choice == '6':
    subject = input("Enter subject to find index: ")
    if subject in schedule:
        print(f"Index of {subject}: {schedule.index(subject)}")
    else:
        print("Subject not found")
elif choice == '7':
    start = int(input("Enter start index: "))
    end = int(input("Enter end index: "))
    print("Sliced tuple:", schedule[start:end])
elif choice == '8':
    print("Sorted Schedule:", tuple(sorted(schedule)))
elif choice == '9':
    subject = input("Enter new subject to add: ")
    temp_list = list(schedule)
    temp_list.append(subject)
    print("temp_list:", temp_list)
    schedule = tuple(temp_list)
```

```
    print("Subject added successfully")
    print("Schedule_tuple:", schedule)

elif choice == '10':
    print("Alphabetically smallest:", min(schedule))
    print("Alphabetically largest:", max(schedule))
elif choice == '11':
    print("Exiting program.")
    break
else:
    print("Invalid choice! Please enter 1-11.")
```

```
# Employee Salary Database - Dictionary Operations
employees = {
    1001: 35000,
    1002: 42000,
    1003: 50000,
    1004: 65000,
    1005: 70000
}
def display_menu():
    print("\n====== Employee Salary Database ======")
    print("1. Display all employee records")
    print("2. Add new employee")
    print("3. Update employee salary")
    print("4. Remove employee")
    print("5. Search employee salary")
    print("6. Display employee IDs")
    print("7. Display salaries")
    print("8. Total number of employees")
    print("9. Clear all records")
    print("10. Exit")
    print("=====================================")
display_menu()
while True:
    choice = int(input("Enter your choice (1-10): "))
    if choice == 1:
        print("\nEmployee Records:")
        for emp_id, salary in employees.items():
            print("Employee ID:", emp_id, "Salary: ₹", salary)
    elif choice == '2':
        emp_id = int(input("Enter employee ID: "))
```

```
    salary = int(input("Enter salary: "))
    employees[emp_id] = salary
    print("Employee added successfully.")
elif choice == '3':
    emp_id = int(input("Enter employee ID to update: "))
    if emp_id in employees:
        salary = int(input("Enter new salary: "))
        employees[emp_id] = salary
        print("Salary updated successfully.")
    else:
        print("Employee not found.")
elif choice == '4':
    emp_id = int(input("Enter employee ID to remove: "))
    if emp_id in employees:
        employees.pop(emp_id)
        print("Employee removed successfully.")
    else:
        print("Employee not found.")
elif choice == '5':
    emp_id = int(input("Enter employee ID to search: "))
    if emp_id in employees:
        print("Salary: ₹", employees.get(emp_id))
    else:
        print("Employee not found.")
elif choice == '6':
    print("Employee IDs:", list(employees.keys()))
elif choice == '7':
    print("Salaries:", list(employees.values()))
elif choice == '8':
    print("Total Employees:", len(employees))
```

```
elif choice == '9':
    employees.clear()
    print("All employee records cleared.")
elif choice == '10':
    print("Exiting Employee Salary Database.")
    break
else:
    print("Invalid choice! Please enter 1-10.")
```

```
# String Operations
message = "  Hello food delivery team, my order was delayed and the food was cold when delivered. Please resolve this issue and refund if possible. Thank you team  "
def display_menu():
    print("\n====== Customer Support Portal for Food Delivery APP======")
    print("1. Enter customer message")
    print("2. Display customer message")
    print("3. Find length of message")
    print("4. Convert message to UPPERCASE")
    print("5. Convert message to lowercase")
    print("6. Convert message to Title Case")
    print("7. Remove extra spaces from message")
    print("8. Replace a word in message")
    print("9. Find position of a word")
    print("10. Count occurrence of a word")
    print("11. Split message into words")
    print("12. Join words using '-'")
    print("13. Check starting word")
    print("14. Check ending word")
    print("15. Slice message")
    print("16. Exit")
    print("========================================")
display_menu()
while True:
     choice = input("Enter your choice (1-16): ")
    if choice == '1':
        print("1. Enter customer message")
        message = input("Enter customer message: ")
        print("Message stored successfully.")
```

```
elif choice == '2':
    print("2. Display customer message")
    print("Customer Message:")
    print(message)

elif choice == '3':
    print("3. Find length of message")
    print("Length of the message is:", len(message))

elif choice == '4':
    print("4. Convert message to UPPERCASE")
    message = message.upper()
    print("Message converted to UPPERCASE:")
    print(message)

elif choice == '5':
    print("5. Convert message to lowercase")
    message = message.lower()
    print("Message converted to lowercase:")
    print(message)

elif choice == '6':
    print("6. Convert message to Title Case")
    message = message.title()
    print("Message converted to Title Case:")
    print(message)

elif choice == '7':
    print("7. Remove extra spaces from message")
    message = message.strip()
```

```
    print("Extra spaces removed:")
    print(message)

elif choice == '8':
    print("8. Replace a word in message")
    old = input("Enter word to replace: ")
    new = input("Enter new word: ")
    message = message.replace(old, new)
    print("Word replaced successfully:")
    print(message)

elif choice == '9':
    print("9. Find position of a word")
    word = input("Enter word to find: ")
    pos = message.find(word)
    if pos != -1:
        print(f"'{word}' found at position {pos}")
    else:
        print("Word not found in the message")

elif choice == '10':
    print("10. Count occurrence of a word")
    word = input("Enter word to count: ")
    count = message.count(word)
    print(f"'{word}' occurs {count} time(s) in the message")

elif choice == '11':
    print("11. Split message into words")
    words = message.split()
    print("Message split into words:")
```

```
    print(words)

elif choice == '12':
    print("12. Join words using '-'")
    words = message.split()
    message = "-".join(words)
    print("Words joined using '-':")
    print(message)
elif choice == '13':
    print("13. Check starting word")
    word = input("Enter starting word: ")
    print("Starts with given word?:", message.startswith(word))
elif choice == '14':
    print("14. Check ending word")
    word = input("Enter ending word: ")
    print("Ends with given word?:", message.endswith(word))
elif choice == '15':
    print("15. Slice message")
    start = int(input("Enter start index: "))
    end = int(input("Enter end index: "))
    print("Sliced message:")
    print(message[start:end])
elif choice == '16':
    print("16. Exit")
    print("Exiting Bank Customer Support Portal.")
    break
else:
    print("Invalid choice! Please enter a valid option.")
```

```
# Bank Employee Training Programs - Set Operations

cyber_security = {101, 102, 103, 104}
data_analytics = {103, 104, 105, 106}

def display_menu():
    print("\n====== Bank Training Programs (Set Operations) ======")
    print("1. Display training participants")
    print("2. Add employee to Cyber Security training")
    print("3. Remove employee from Cyber Security training")
    print("4. Employees attending both trainings (Intersection)")
    print("5. Employees attending any training (Union)")
    print("6. Employees only in Cyber Security training (Difference)")
    print("7. Employees in exactly one training (Symmetric Difference)")
    print("8. Check subset and superset")
    print("9. Total participants in each training")
    print("10. Clear all training records")
    print("11. Exit")
    print("=====================================================")
display_menu()
while True:

    choice = input("Enter your choice (1-11): ")

    if choice == '1':
        print("1. Display training participants")
        print("Cyber Security Training:", cyber_security)
        print("Data Analytics Training:", data_analytics)

    elif choice == '2':
        print("2. Add employee to Cyber Security training")
        emp_id = int(input("Enter employee ID to add: "))
        cyber_security.add(emp_id)
        print("Employee added to Cyber Security training.")

    elif choice == '3':
        print("3. Remove employee from Cyber Security training")
        emp_id = int(input("Enter employee ID to remove: "))
        cyber_security.discard(emp_id)
        print("Employee removed (if present).")

    elif choice == '4':
        print("4. Employees attending both trainings (Intersection)")
```

```
        print("Employees attending both trainings:",
              cyber_security.intersection(data_analytics))

    elif choice == '5':
        print("5. Employees attending any training (Union)")
        print("Employees attending any training:",
cyber_security.union(data_analytics))

    elif choice == '6':
        print("6. Employees only in Cyber Security training (Difference)")
        print("Only Cyber Security training:",
cyber_security.difference(data_analytics))

    elif choice == '7':
        print("7. Employees in exactly one training (Symmetric Difference)")
        print("Exactly one training:",
cyber_security.symmetric_difference(data_analytics))

    elif choice == '8':
        print("8. Check subset and superset")
        print("Is Cyber Security subset of Data Analytics?",
cyber_security.issubset(data_analytics))
        print("Is Cyber Security superset of Data Analytics?",
cyber_security.issuperset(data_analytics))

    elif choice == '9':
        print("9. Total participants in each training")
        print("Cyber Security participants:", len(cyber_security))
        print("Data Analytics participants:", len(data_analytics))

    elif choice == '10':
        print("10. Clear all training records")
        cyber_security.clear()
        data_analytics.clear()
        print("All training records cleared.")

    elif choice == '11':
        print("11. Exit")
        print("Exiting Bank Training System.")
        break

    else:
        print("Invalid choice! Please enter 1-11.")
```

Implement Method Overriding

Problem Statement:

Write a python program to implement a shopping system where different item categories use method overriding to apply their own specific discount rates.

Coding:

```
# Base Class
class Item:
    def __init__(self, name, price):
        self.name = name
        self.price = price

    def apply_discount(self):
        return self.price  # No discount by default

# Subclass for Electronics
class Electronics(Item):
    def __init__(self, name, price, warranty_years):
        super().__init__(name, price)
        self.warranty_years = warranty_years

    def apply_discount(self):
        # Electronics get a 10% discount
        discount = 0.10
        return self.price * (1 - discount)

# Subclass for Clothing
class Clothing(Item):
    def __init__(self, name, price, size):
        super().__init__(name, price)
        self.size = size

    def apply_discount(self):
        # Clothing gets a 20% discount
        discount = 0.20
        return self.price * (1 - discount)

# Function to demonstrate method overriding
def main():
    # Create and describe items
    electronic_item = Electronics(name="Laptop", price=1000, warranty_years=2)
    clothing_item = Clothing(name="T-Shirt", price=50, size="M")

    # Print discounted prices
    print(f"{electronic_item.name} discounted price: Rs. {electronic_item.apply_discount():.2f}")
    print(f"{clothing_item.name} discounted price:Rs. {clothing_item.apply_discount():.2f}")

# Run the main function
main()
```